Q1 For the below device I-V plot: Calculate approximately.

a. How much change in voltage required to change the current by one decade (10 times) i.e. $SS = \frac{dV}{d(\log_{10} I)}$ (mV/dec). Consider voltage range $\leq$ 0.8V where SS is valid. **(2)**

b. How much is the leakage if measured at 0.2V? **(1)**

c. How much is the leakage at 0.2V if $SS = 200\ mV/dec$. Assume same current given in plot at 0.8V **(2)**

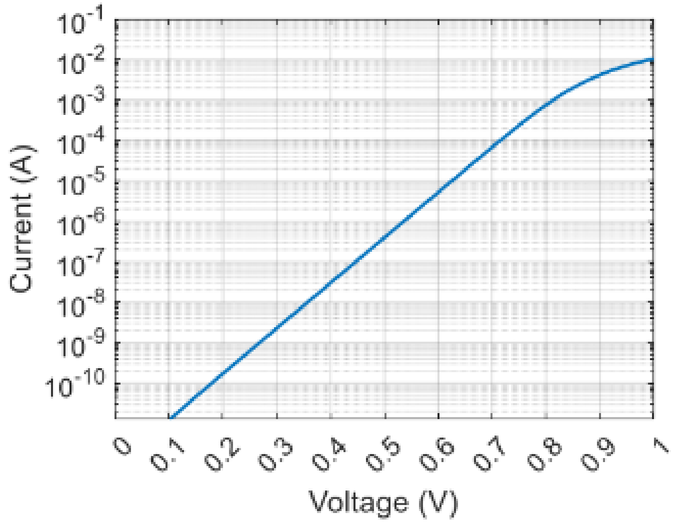

Figure 1: Current vs Voltage Characteristics

Solution:

(a) Validity for 0.8V.

We know that Subthreshold Swing (SS) is defined as:

$$SS = \frac{dV}{d(\log_{10} I)}$$

At 0.8V, using values from the graph:

$$\begin{aligned} SS &= \frac{0.8 - 0.2}{\log_{10}(10^{-4}) - \log_{10}(10^{-10})} \\ &= \frac{0.6}{\log_{10}(10^{6})} \\ &= \frac{0.6}{6}\,\frac{\text{V}}{\text{dec}} \\ &= 0.1\,\frac{\text{V}}{\text{dec}} \\ &= \mathbf{100}\,\frac{\text{mV}}{\text{dec}} \end{aligned}$$

Alternately,

SS valid ≤ 0.8 V

@ $I = 10^{-4}$ A, $V_1 \simeq 0.72$ V

@ $I = 10^{-5}$ A, $V_2 = 0.62$ V

$\Delta V = V_1 - V_2 \simeq 100$ mV/dec

(b) Given $SS = 100\,\text{mV/dec}$. We need to find the current $I_{leakage}$.

$$\begin{aligned}
SS = \frac{100\text{mV}}{\text{dec}} &= \frac{0.8 - 0.2}{\log_{10}(10^{-3}) - \log_{10}(I)} \\
0.1 &= \frac{0.6}{\log_{10}\left(\frac{10^{-3}}{I}\right)} \\
\log_{10}\left(\frac{10^{-3}}{I}\right) &= \frac{0.6}{0.1} \\
\log_{10}\left(\frac{10^{-3}}{I}\right) &= 6 \\
\frac{10^{-3}}{I} &= 10^6 \\
I &= \frac{10^{-3}}{10^6} \\
\mathbf{I} &= \mathbf{10^{-9}}\,\text{A}
\end{aligned}$$

@ $V = 0.2V$,

$I \approx 10^{-9} - 10^{-10}$ A

"range"

(c) The leakage at 0.2V Given $SS = 200\,\text{mV/dec}$.
We know,

$$SS = \frac{dV}{d(\log_{10} I)}$$

Where V is voltage and I is current.
From the graph, we can see current at 0.8V is 10^{-3} A. To find current (leakage) at 0.2V, let the current be I_1.
Given $SS = 200\,\text{mV/dec} = 0.2\,\text{V/dec}$.
Substituting into the formula:

$$\begin{aligned}
0.2 &= \frac{0.8 - 0.2}{\log_{10}(10^{-3}) - \log_{10}(I_1)} \\
0.2 &= \frac{0.6}{\log_{10}(10^{-3}) - \log_{10}(I_1)} \\
\log_{10}(10^{-3}) - \log_{10}(I_1) &= \frac{0.6}{0.2} \\
-3 - \log_{10}(I_1) &= 3 \\
-\log_{10}(I_1) &= 3 + 3 \\
-\log_{10}(I_1) &= 6 \\
\log_{10}(I_1) &= -6 \\
\log_{10}(I_1) &= \log_{10}(10^{-6})
\end{aligned}$$

@ $0.8V$, $I = 10^{-3}$A

with 0.2 V/de

$\Rightarrow \frac{0.8 - 0.2}{0.2}$

3 order change

$\frac{10^{-3}}{10^3} \Rightarrow 10^{-6}$ A

Thus, the current I_1 is:

$$\mathbf{I_1 = 10^{-6}}\,\text{A}$$

The leakage current at 0.2V with $SS = 200\,\text{mV/dec}$ is 10^{-6} A.

Q2. For Silicon (Diamond crystal structure) with the lattice constant of 5.43 Å.

a. Find volume density (#/cm^3) **(2)**

b. Calculate atomic packing fraction (APF) **(1)**

Q3. For the article "Plenty of Room at the Bottom" (Write answers in 1-2 lines only.)

a. What is the core message from the article "Plenty of Room at the Bottom"? **(1)**

b. What are the different methods suggested to write and read on the tiny (few atoms) scale? **(1)**

Solution:

2. (a) Volume Density

For a Diamond crystal structure (like Silicon), the number of effective atoms per unit cell (N) is 8.

Given lattice constant $a = 5.43\,\text{Å} = 5.43 \times 10^{-8}\,\text{cm}$.

The volume density (n) is given by:

$$n = \frac{\text{Number of atoms}}{\text{Volume of unit cell}} = \frac{N}{a^3}$$

Substituting the values:

$$n = \frac{8}{(5.43 \times 10^{-8}\,\text{cm})^3}$$
$$n = \frac{8}{160.103 \times 10^{-24}\,\text{cm}^3}$$
$$n = 0.04996 \times 10^{24}\,\text{cm}^{-3}$$
$$\mathbf{n \approx 5 \times 10^{22}}\,\text{atoms/cm}^{\mathbf{3}}$$

(b) Atomic Packing Fraction (APF)

The APF is defined as the ratio of the volume occupied by the atoms to the total volume of the unit cell.

$$\text{APF} = \frac{N \times \text{Volume of one atom}}{a^3}$$

In a diamond cubic structure, the relationship between the lattice constant a and the atomic radius r is:

$$8r = \sqrt{3}a \quad \implies \quad r = \frac{\sqrt{3}}{8}a$$

Substituting this into the APF formula:

$$\begin{aligned}
\text{APF} &= \frac{8 \times \frac{4}{3}\pi r^3}{a^3} \\
&= \frac{32\pi}{3a^3}\left(\frac{\sqrt{3}a}{8}\right)^3 \\
&= \frac{32\pi}{3a^3} \cdot \frac{3\sqrt{3}a^3}{512} \\
&= \frac{\sqrt{3}\pi}{16} \\
&= \frac{1.732 \times 3.1415}{16} \\
\text{APF} &\approx \mathbf{0.34}
\end{aligned}$$

Question 3: Solution

(a) The core message is that the laws of physics do not prohibit the manipulation of things on a small scale, specifically atom-by-atom. Feynman argues that there is ”plenty of room” to decrease the size of things for practical applications—such as storing vast amounts of information or manufacturing tiny machines without violating any fundamental physical principles.

(b) Writing:

Reverse Electron Microscope: Using the lenses of an electron microscope in reverse to demagnify a beam of ions to a very small spot, allowing one to ”write” by depositing material in lines (similar to a TV cathode ray oscilloscope).

Electron Beam via Light: Using an optical microscope backwards to focus light onto a photoelectric screen, which releases electrons that are then focused down by electron microscope lenses to etch or modify a metal surface.

Reading:

Plastic Mold Replica: Pressing the tiny written metal text into plastic to make a mold, evaporating silica into it, shadowing it with gold, and then viewing the thin film through an electron microscope.

Direct Observation: Improving electron microscopes by 100 times to simply ”look at the thing” and see individual atoms directly.